Supplementary Material: Retrieval-Assisted Scalar Grounding for Natural-Language Optimization: A Diagnostic Study

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This supplement preserves secondary evidence, population boundaries and source provenance. It accompanies the main article and includes the archived 217-relationship Gemini–DeepSeek–Cohere audit, with explicit recovery and denominator accounting. No new model predictions or experiments were produced for M5.

S1 System Context and Secondary Protocol Tables

Table S1 Task-level comparison landscape for recent natural-language optimization modeling systems. This table is deliberately *not* a leaderboard: systems differ in output type, supervision, and evaluation protocol, so accuracy numbers are omitted. Abbreviations: Ret. = retrieval/memory; Sol. = solver feedback during generation or training; FT = fine-tuning or RL training; Comp. = directly comparable to the InstantiationReady / Exact20 (on hits) protocol of this paper.

Method	Year	Task	Output	Ret.	Sol.	FT	Comp.
OptiMUS	2024	End-to-end (MI)LP	Model+code	no	yes	no	no
OptLLM	2024	NL→model/code	Model+code	no	yes	no*	no
Chain-of-Experts	2024	Multi-agent OR	Model+code	no	yes	no	no
LLMOPT	2025	Fine-tuned modeling	Model+code	no	yes	yes	no
OPT2CODE	2025	Retrieval+codegen	Solver code	yes	yes	no	no
ORLM	2025	Training+feedback	Model+code	no	yes	yes	no
OptMATH	2025	Data synthesis	Model+code	no	yes	yes	no
SIRL	2025	Solver-informed RL	Model+code	no	yes	yes	no
OR-R1	2026	Test-time RL	Model+code	no	yes	yes	no
DeepOR	2026	Foundation OR	Formulation	—	yes	yes	no
OptimAI	2025	Multi-agent NL-OR	Model+code	no	yes	no	no
AlphaOPT	2025	Experience library	Model+code	yes	yes	no‡	no
ORThought	2026	Logistics CoT	Model+code	no	yes	no	no
Ours	2026	Fixed-catalog inst.	Schema+scalars	yes	no	no	yes

*OptLLM supports both prompting and fine-tuned variants. ‡Library learning without weight updates. "—" denotes information not established from a verified public artifact at the time of writing. Direct comparability (Comp.) is "yes" only for systems that share our fixed-catalog scalar-instantiation task and InstantiationReady / Exact20 protocol; end-to-end solver accuracy is a different metric.

Table S2 Retrieval methods used for schema identification.

Method	Representation	Role in pipeline
BM25	Probabilistic lexical weighting	Sparse lexical baseline
TF–IDF	Weighted bag-of-words cosine similarity	Primary lexical baseline
LSA	Latent projection of TF–IDF features	Lower-dimensional lexical-semantic baseline
BGE-M3	Pretrained dense text embeddings	Modern semantic retrieval baseline

Table S3 Summary of the experimental blocks used in the paper.

Block	Primary purpose	Main outputs
Main benchmark	Retrieval and scalar grounding on NLP4LP	Schema R@1, Coverage, TypeMatch, Exact20 (on hits), InstantiationReady
Robustness analyses	Sensitivity to degraded input and lexical overlap	Orig / noisy / short retrieval comparisons, overlap-based stress tests
Numeric-extraction ablation	Isolate effect of corrected ratio-word extraction	TypeMatch and InstantiationReady changes (baseline vs. ratio-aware extraction)
Retrospective paired analysis	Quantify observed paired differences	Group-bootstrap intervals, exact swap tests and Holm adjustment
Proxy checks (60 inst.)	Coverage/type and rendered-template checks	Historical proxy and template-completion rates
Extended proxy checks (269 inst.)	Same lightweight checks on a code-available subset	Schema hit, proxy and template-completion rates
Restricted solver-backed execution (20 inst.)	Reported reference-program execution via SciPy HiGHS shim	Executable, solver-success, feasible, and objective-produced rates

S2 Secondary Retrieval and Restricted Execution Diagnostics

Table S4 Historical sanitized-retrieval aggregates (*Schema R@1*; 331 queries per cell). The 331-document diagnostic index uses 256-dimensional LSA and differs from the 335-document index in Table 1. Per-condition predictions are unavailable; agreement with the retained aggregate is not prediction-level verification.

Sanitization	TF–IDF	BM25	LSA
None (baseline)	0.9063	0.8852	0.8459
Numbers removed	0.9063	0.8973	0.8459
Stopwords removed	0.9124	0.9063	0.9184
Numbers & stopwords removed	0.9124	0.9063	0.9184

Table S5 Historically reported 60-case proxy checks using optimization-role relation repair; per-instance outputs are unavailable for reconstruction. All rates divide by 60. Proxy and template-completion predicates are defined in the text; neither checks numerical correctness.

Baseline	Schema Hit	Coverage/type proxy	Template complete
TF–IDF	0.9333	0.7500	0.7500
BM25	0.9000	0.7333	0.7333
Oracle	1.0000	0.7667	0.7833

Table S6 Historically reported gold-assisted execution on 20 instances (SciPy HiGHS MILP shim); per-instance outputs and the original manifest are unavailable for reconstruction. Every rate divides by 20.

Baseline	Executable	Solver Success	Feasible	Objective Produced
TF–IDF	0.95	0.80	0.80	0.80
Oracle	0.95	0.75	0.75	0.75

Table S7 One-to-one factorial controls on the same 49 queries and 277 gold scalar slots. Entries show greedy → exact maximum-sum assignment. E/T denote numerical exact/tolerance-correct counts. Full instances also require compatible types and every gold scalar key. These controls are distinct from the historical 331-query configurations.

Pool	Scorer	E slots	T slots	E full	T full
Typed	Ordinal	71 → 71	87 → 86	3 → 3	3 → 3
Typed	Context	106 → 136	117 → 145	3 → 4	5 → 5
Typed	No marker bonus	97 → 137	111 → 146	3 → 4	5 → 5
Rich	Ordinal	67 → 67	81 → 81	3 → 3	3 → 3
Rich	Context	100 → 133	111 → 142	3 → 4	5 → 5
Rich	No marker bonus	97 → 135	111 → 144	3 → 4	5 → 5

S3 Cached Grounding, References and Reproduction

S3.1 Full Factorial and Search-Retention Evidence

Section 5.4 distinguishes the historical 40-versus-7 all-key result from the controlled 49-query study. The latter's 277 scalar slots and reference-derived template inventory are fixed within each decoder comparison. The known excluded query shares a delexicalized text group with earlier training/development data; grouping uses transitive five-gram Jaccard similarity at least 0.80. Exclusion addresses that known crossing, not prior benchmark exposure or unrecognized semantic duplicates.

The design crosses two candidate constructors, three scorers, two decoders and two mention-reuse policies (24 cells). Decoder pairs share the exact same edge-score matrix, canonical lexical slot order, and zero-score null option. Scorers are an explicit ordinal encoding of the original type preference, the frozen contextual scorer, and a subtraction of only its percent/currency marker bonuses. The ordinal control introduces a documented cardinalization and does not reproduce every historical assignment. One-to-one assignment is a diagnostic restriction rather than a physical law. With unrestricted reuse and additive scores, independent per-slot maximization is already globally optimal; every corresponding local/global output agrees. Table S7 shows the one-to-one controls; all 24 cells, candidate identities, matrices and per-query predictions are retained in the reproducibility artifacts.

With contextual scores, exact assignment increases exact numerical slot counts by 30/277 (typed pool) and 33/277 (rich pool). Existing paired query-group bootstrap intervals ($B = 2000$, seed 42) are respectively $[3.14, 17.95]$ and $[4.07, 18.84]$ percentage points. These are marginal, exploratory intervals. The full-tolerance count remains 5/49 for both decoders, with one gained and one lost query; the paired interval is $[-6.12, 6.12]$ percentage points. A separate retrospective exact groupwise method-swap analysis covers every one-factor contrast and four numerical endpoints, using a single Holm family of 240 tests [2]. Neither contextual-score decoder contrast survives that familywise adjustment (adjusted $p = 1$). Some contextual-versus-ordinal slot comparisons survive; no whole-instance endpoint does. These results support descriptive

component differences, not a confirmed whole-instance improvement or attribution of all 33 historical additional successes to search. The recorded sub-millisecond cell timings include metric and instrumentation overhead and are not clean algorithm-speed comparisons.

Candidate availability, pruning and objective quality. The earlier 50-query joint-grounding pilot selects 3 tolerance-correct instances with either local or joint scoring. In the saved common pools, 32/50 permit a criterion-correct complete mapping, but only 8/50 retain one in the top ten. Reconciliation with identical saved beam outputs separates 14 cases losing correct prefixes, 6 generating then pruning a complete correct mapping, 4 losing it at the final top-ten cut, 5 retaining it below the selected mapping, and 3 selecting it. The other 18 cases lack a complete compatible mapping in the pool or inventory. Width and ranking limitations must therefore be distinguished from candidate absence. Optimality claims about beam methods also depend on their search restriction [4]; that neural-generation result is methodological context, not a guarantee for this grounding algorithm.

An exact MILP diagnostic holds the cached local/pairwise objective and domains fixed. Among the 32 pool-compatible cases, 23 have a strictly higher-scoring incorrect optimum than their best criterion-correct mapping; 6 have tied correct and incorrect optima, and 3 select a correct optimum. The correctness-restricted solves are oracle diagnostics, never inference procedures. Exact mention-based top-ten enumeration retains a correct mapping in 9/50 cases versus 8/50 for the beam, with tied-optimum membership qualifications. More search cannot repair the 23 strict objective failures. Ten of the 50 exact top-ten lists contain duplicate numerical/type mappings realized by different mention indices. A separate 49-query follow-up ranks without evaluation values, conditional on the same supplied inventory. It reuses 40 already-distinct lists and performs 23 additional solves for the other nine. Every retained top-ten list is certified, but criterion-correct list retention remains 9/49, with zero gained or lost queries; selected correctness remains 3/49. Thus reclaiming this alias budget supplies no observed correctness gain and no justification for full-dataset expansion. Mention equality, numerical equality and semantic identity are different concepts. No gold value is used to rank inference candidates.

Detailed source/reference conventions, the versioned correction overlay and the historical tolerance limitation are given in Appendix S3.2. No reference correction is used during inference, and no new hard unit or mathematical relation was validated by these studies.

S3.2 Reference Conventions and Correction-Only Reanalysis

All 12 previously flagged percentage slots were inspected against the original query, parameter descriptions, schema equations and reference-code annotations. A separate versioned overlay, `numerical_reference_v2`, contains ten fractional-reference corrections (Table S8). The literal percentage and a direct share-coefficient equation establish each conversion: a share p in $X \geq p(X + Y)$ or its rearrangement $(1 - p)X \geq pY$ must use the fraction, not percentage points. Upper bounds reverse the inequality. These corrections do not alter original reference files, any historical measurements,

or inference predictions. They establish a source/schema-consistent convention, not dataset-maintainer approval or correctness of every other constraint in an instance.

Table S8 Complete fractional-reference change inventory. IDs abbreviate `nlp4lp_test_ID`; all values are scalar. Every entry has full source text, a literal span, equation/code witnesses, hashes and a confirmed evidence classification in the overlay.

ID	Parameter	Original	Fractional reference
48	MinLycheePercentage	40	0.40
78	MinChickenCurryPercentage	25	0.25
102	MinPercentageAdultVaccines	70	0.70
157	MaxRickshawPercentage	40	0.40
170	MinBoatTripPercentage	50	0.50
174	MaxPercentageLabradors	60	0.60
177	MaxCompanyCarRidePercentage	60	0.60
180	MinCarTripPercentage	30	0.30
187	MaxVanPercentage	30	0.30
215	MaxTrolleyTransportPercentage	40	0.40

The two remaining flags are the two snack-mix parameters in query 206. They retain 20 and 35 as percentage points: the complementary-ingredient constraint explicitly uses $(100 - p)/100$. Converting those references alone would break that equation. Instead, a separate formulation overlay inserts the missing divisions in the other ingredient constraint:

$$(p_1/100)x_1 + (p_2/100)x_2 \leq b_{\text{cat}}, \tag{S1}$$

where $p_1 = 20$, $p_2 = 35$, x_1, x_2 are mix masses in kilograms, and $b_{\text{cat}} = 20$ kilograms is the available cat-paw ingredient. This is an algebraic/dimensional correction only; no model is executed. Thus a percent-named parameter does not imply a universal fractional reference representation. Query 102 additionally has a metadata-key alias, and query 177 a pluralized coefficient identifier in another constraint; the numerical overlay does not repair those independent schema defects.

Reanalysis is restricted to predictions actually saved. None of the ten changed-reference cases is in the 49-query primary factorial, so all of its metrics and the 240-test family are unchanged. Only development case 177 has affected factorial predictions; the existing version-1 sensitivity is reusable because its correction is unchanged. It adds no fully correct development instance. Reference-only cases without predictions are not run.

For the distinct, pre-ratio Stage-A cache, saved selections over all 331 queries and 1,847 gold scalar slots permit a separate correction-only comparison: numerical exact slots change from 348 to 356, tolerance-correct slots from 401 to 409, and full type-compatible tolerance instances remain 6/331. Missing saved selections count wrong. These are not replacements for the headline typed-greedy 7/331 or constrained 40/331 results, whose historical per-slot values are unavailable. The Stage-A counts use actual saved selections; an earlier metadata-filtered error partition counted 394 correct slots

because seven otherwise correct saved selections lack slots in the later metadata-only snapshot. The inventory difference is explicitly reconciled, rather than described as an experimental change.

The historical numerical tolerance remains $|\hat{y} - y| / \max(1, |y|) \leq 0.20$: for fractions it is absolute 0.20, so prediction 0.40 passes against reference 0.25. It is not uniformly a 20% relative tolerance. The pilot's numerical exact criterion uses the same scaling with threshold 10^{-9}, not bitwise equality. Slot-level numeric correctness does not impose a type gate; full-instance correctness also requires the schema, all gold keys and compatible types. Candidate-pool coverage instead asks whether any permitted option could satisfy a criterion, without claiming that inference retained or selected it. No new tolerance is chosen and no gold value is used during selection.

S3.3 Populations, Evidence and Reproduction Boundaries

The completed archival investigation found no authenticated headline scalar-assignment export. The four existing metric CSVs byte-match their first committed versions; this establishes stored-record integrity, not independent reevaluation of the values originally selected. The inspected original runners serialized 13 metric columns and omitted scalar mappings. Their absence is not evidence of deletion, and source code alone cannot reconstruct a historical prediction.

We distinguish five evidence types. *Original aggregates/query metrics* support summary-statistic reconstruction but not recovery of scalar choices. *Preserved predictions* include top-1 schema IDs, Stage-A selections and later cached mappings. *Deterministically reconstructed statistics* include residual partitions and paired tests derived from those existing records. *Historical diagnostic examples* document selected cases or separate replay configurations; they are not complete original samples. *Unmatched archival candidates* have no authenticated run identity and are excluded from evaluation. No newer outputs replace missing predictions.

The Stage-A cache covers 311 queries with 1,705 slot rows; its separate full-frame sensitivity uses 331 queries and 1,847 gold scalars, counting missing selections as missing. The initial factorial collection contains 49 primary, 12 development, one overlap-reconciliation and two reference-only cases. All 1,176 primary method/query choices were reevaluated from saved indices/candidate values with historical type rules; no decoder or solver was rerun.

The headline per-query files are under `results/selective_grounding_rerank/`, `results/final_resubmission_method/`, `results/dense_retrieval_bge_m3/` and `results/oracle_recomputation_2026-08-15/`. Matched alternatives are under `results/stage6_matched_grounding_baselines_2026-08-27/`; joint traces are under `results/joint_grounding_2026-10-27/`. Original factorial and enumeration checkpoints are under `results/wulver_grounding_attribution_2026-10-27/slurm_1462366/` and `results/grounding_followup_2026-10-27/slurm_1462557/`.

Saved top-1 IDs reconstruct retrieval rates; saved query metrics reconstruct the main diagnostic tables and paired statistics (10,000 group resamples, NumPy seed 42). The corrected TF–IDF variant mean is 0.8640, computed before rounding from

Table S9 Evaluation populations and preserved evidence. These populations and configuration versions must not be interchanged.

Evidence	Unit and denominator	Available observations
Retrieval	331 queries per variant	Top-1 schema IDs
Main scalar diagnostics	331 queries per arm	Query metrics, not slots
Conditional Exact20	291/291/303/320 queries	Per-query fractions
Pre-ratio Stage A	1,705 rows / 311 queries	Mentions and selections
Joint pilot	50 queries; 280 gold scalars	300 method/query traces
Primary factorial	49 queries; 277 gold scalars	Choices and matrices
Initial factorial collection	64 cases; mixed populations	64 checkpoints
Distinct-class enumeration	49 primary queries	49 saved checkpoints
Subset proxy/solver checks	60/269/20 filtered queries	Reports/aggregates
External generation	Same 18 source problems	Artifacts/execution logs
External extraction	1,000 instances	Extracted sets and counts
Relationship QC	217 items / 16 problems	651 final objects and recovery lineage

$(301+299+258)/(3\cdot331)$. Main scalar files cannot support corrected-reference reevaluation. The BGE reporting generator writes literal metadata/aggregates, not a runtime receipt: its revision field and inferred device do not authenticate executed weights, environment or hash-dependent slot order.

The sanitized-retrieval and 60/269/20 complete results remain supported by historical aggregates/reports only. Five reported examples each from 60 and 269 include three shared query IDs, but do not recover either full manifest. The 20-case outcome writer itself omits full scalar maps. Imported overlap-study hit flags, metadata-order diagnostic replays and unauthenticated loose blobs cannot substitute for the missing target outputs.

External generation preserves artifacts and execution records, but incompletely retained prompts prevent authentication of every historical input. Gold objectives are scoring-only in inspected generation paths; objective agreement is not formulation equivalence. The extraction audit preserves automatic sets/counts. Its later corrected aggregate ratio is checkable, but missing corrected per-instance counts prevent independently reconstructing its historical bootstrap interval.

The M3 correction uses `tools/reaggregate_common18_m3.py` and the pure `baselines/comparison/objective_evidence.py` module. From a checkout containing the M3 correction scripts, run `python` with script argument `tools/reaggregate_`

`common18_m3.py`, `--output` and a separate local audit directory. The command reads saved records only and never overwrites historical results. Independent verification and regression instructions accompany the local correction ledger under `docs/manuscript_m3_2026-10-28/`. This ledger identifies every case, source hash, historical/corrected disposition and precision exclusion.

The M5 local release candidate supplies a machine-readable file inventory, hashes, excluded-file reasons and offline commands in `reproducibility/m5/README.md`. Its deterministic entry point, `tools/verify_llm217_m5.py`, reconstructs judgments from frozen HTTP responses without importing historical validators or model SDKs. The 651-row judgment ledger and the 20 invalid-case IDs are derived exports; originals are preserved. Account/project identifiers are removed only from explicitly marked public copies, with separate original and release hashes. No raw NLP4LP cache or OptMATH problem corpus is redistributed. The already public derived-metrics dataset keeps its existing CC BY-NC-SA 4.0 license; NL4Opt source material keeps its MIT notice. M5 is a local author-review candidate, not an uploaded release.

Direct offline checks reproduce retained metric aggregates, corrected common-18 text measurements and relationship validity/agreement. Fresh source-level evaluation requires separately obtained upstream data and appropriate dependencies; it was not run here. Historical aggregates without membership records and missing original headline scalar maps remain verifiable only to their retained evidence boundaries. No reconstructed output fills these gaps.

S3.4 Input Dependencies and Identifiable Conclusions

For the headline scalar criterion, let F be the set passing schema match, full eligible coverage, all types and full numerical tolerance; let S and I pass Strict and ordinary coarse readiness. By their definitions, $F \subseteq S \subseteq I$: full coverage/type gates imply the 0.8 gates, and Strict adds schema match. Thus the saved ratio-aware counts satisfy $7 \leq 255 \leq 265$ on the same 331-query frame. This is a logical relation between conditional diagnostics, not a bound on autonomous or semantic accuracy. Without assumptions linking a system supplied with a reference inventory to a system that must infer that inventory, the latter's success rate is unidentified. Nor does stored-reference tolerance certify source semantics when the references themselves contain errors.

The absence of new fitting in this revision does not establish evaluation independence. The earlier learned train/development/test subsets all come from the 331-query evaluation-origin corpus; retained grouping records identify train/development and development/test crossings. The 49-case primary slice excludes the known query-291 crossing, but does not certify new template families or a fresh test set. The separate 399-problem NL4Opt quarantine is not an NLP4LP split and was not applied to these tables.

S3.5 Historical Supervision Experiments: Evidence and Limits

Completed training, negative performance, inconclusive evidence and unavailable supervision are different outcomes. Retained training logs establish that a DistilRoBERTa mention–slot classifier was actually fitted, rather than merely implemented.

Table S10 Verified input boundaries. "Reference-derived" describes provenance even when a cached template is treated as a supplied catalog.

Stage	Prediction-time inputs	Reference use / interpretation
Retrieval	Query; reference-derived catalog descriptions	Partially assisted known-template identification; target ID used only for scoring, except Oracle
Main grounding	Query; selected template keys; query gold scalar eligibility	Gold-assisted assignment; gold magnitudes used only for scoring
Joint/factorial controls	Cached mentions; template names, shapes, descriptions and available relations; name-inferred types	Partially assisted; no query gold-key filter or value-guided inference ranking
Reference execution	Selected reference program; default parameters; predicted scalar replacements	Gold-assisted; structured and unfilled values retained
External generators	Problem text; fixed prompt; model; own solver feedback where configured	Inspected prompts exclude reference targets; gold objectives score outputs

Its weak labels selected the first numerically matching mention. The largest retained run (job 854626) used 230/50/50 internal train/development/test instances drawn from the existing NLP4LP evaluation corpus. The logged selection counts are 51/280 slots and 0/45 legacy exact-instance outcomes. Five nominal test instances do not enter that latter denominator. Those metrics are not the all-required-scalar criterion used in this manuscript.

A later summary's claim that this run was worse than a rule baseline on every metric conflicts with the preserved stdout. The audit also identifies an evaluator path that can instantiate an untrained neural head for a nominal rule configuration. Missing original run checkpoints, exact run source SHAs and prediction traces prevent resolving that comparison by replay. Therefore these logs demonstrate completed weak-label learning with poor recorded absolute outcomes, but not a clean causal comparison of neural learning against rules. No headline result is substituted from the conflicting summary.

P0 is a separate completed classifier-plus-assignment study. Logistic regression over 25 handcrafted features and one frozen MiniLM cosine feature was selected on development slot selection; the encoder was not fine-tuned. On its 50 oracle-schema cases, independent selection has coarse readiness 0.80, exact maximum-weight assignment 0.78, and assignment plus repair 0.78, versus a separately retained typed-greedy comparator at 0.86. These are the saved readiness outcomes, not whole-instance numerical correctness. The probability outputs have not been shown calibrated; the exact assignment has no private null option, and several components differ from the comparator. Its first-value-match labels and evaluation-origin split do not establish an independent semantic test.

Both histories used internal subsets of the 331-query corpus; disjoint example IDs alone did not establish independence. The saved text-group audit finds train/development and development/test overlap. This compromises any later use of those fits as independent evidence on the full 331 queries. It does not retroactively turn every internal observation into a nonexistent experiment. The defensible lesson is narrower: fitting an equality-derived target and optimizing its scores did not establish trustworthy occurrence-specific supervision.

MiniLM cross-encoder and multitask grounder implementations also exist, but a completed non-smoke run for the former and a completed NL4Opt auxiliary neural checkpoint for the latter were not verified. They are not counted as negative model results. Actual GAMS and targeted-synthetic auxiliary runs used only two and one effective test problems; they are too small and weakly supervised to justify general transfer claims. Those execution records remain in the audit inventory rather than a new performance table. Fine-tuned retrieval addresses a different task, and an encoder substitution alone is not a new relationship-grounding contribution.

S3.6 NL4Opt Relationship and Independence Audit

The audit reads 713 NL4Opt source problems at pinned revision `49f1e0d66b7fdcd33305a7f281c2a7c13f5620ea`. Its 3,114 positive proposals and 3,246 ambiguous links are derived occurrence–declaration candidates. They do not constitute new independent human annotations. A conservative source-template investigation propagates quarantine to 399 problems; 314 remain unflagged but uncertified. Earlier lexical-screen negatives and later structural-template concerns are compatible because they test different relations. Quarantine counts are precautionary grouping outcomes, not measured contamination or semantic-error rates.

A separate relationship-labeling recovery audit is discussed in the main paper. Its numerical provider comparison, recovery protocol and complete-case denominators appear in main Section 6.3 and Supplement, Section S6.1. No machine consensus has been accepted as a human semantic reference, and no model is trained from that packet for this revision.

S4 LLM Formulation Provenance and Individual Outcomes

S4.1 Implementation Provenance and Corrected Predicates

All four evaluated systems attempt the same 18 source problem numbers, selected before the comparison by the fixed `comparison-nlp4lp-common-18-v1` manifest. These numbers map to zero-based NLP4LP query IDs; retained input hashes agree for every Python system. Full rendered prompt retention is incomplete. No model inference or solver execution is repeated in this correction.

The Azure responses identify `gpt-5.4-2026-03-05`. OptMATH checkpoint revision is `617fe77`; upstream code is `f15bbc4477c70db85ad148df8bcc1b780bca0f8c`. ORLM checkpoint revision is `94fdc3c5738c6536d4880dc19a78f215529181c5`;

Table S11 External-baseline provenance on the common-18 subset. The PaMOP row is a project reconstruction; adapted rows use released checkpoints with project-maintained interfaces. Ours is listed for context only; native metrics are not comparable to Table 8.

System	Provenance	Output representation	Solver environment / status	Cases
Ours (this paper)	NATIVE	Scalar slot assignments on retrieved schema	No solver call in core benchmark	18
PaMOP [6]	Project reconstruction	AMPL formulation + solver code	Local AMPL/Gurobi; 13/18 successful	18
ORLM [3]	ADAPTED-OFFICIAL	LP formulation + solver code	Solver dependency (`coptpy`) unavailable; execution blocked	18
OptMATH [5]	ADAPTED-OFFICIAL	LP formulation + solver code	Gurobi (`gurobipy`); 15/18 executed	18
Generic LLM (GPT-5.4-2026-03-05)	GENERIC-ZERO-SHOT	Gurobi solver code, zero-shot	Gurobi (`gurobipy`); 16/18 executed	18
DeepOR [7]	—	Formulation (paper-described)	No evaluable artifact located in the August 2026 audit	0
OR-R1 [1]	—	Formulation + solver code	Official code found; checkpoint unavailable in the August 2026 audit	0

upstream code is `33bc47d0a1d1710d24ab839118bdf4cb89b9e31b`. Generation adaptations identify local `69df7c030fabb3bd785684b68a844929487603b9`, except ORLM at `1831396b0b3d4428415e354b0a4e1fcbc658df26`. The later direct-Gurobi execution script is at `0197f0cf737f162bd65128c532c708177f475e33`; it is absent from the earlier generation commit. These stages have separate provenance. Successful logs name Gurobi 13.0.2; 12 PaMOP successes name that solver and one is solved by AMPL presolve. These records do not certify every environment default or hosted weight byte.

The corrected Python comparison uses the original round-first predicate in main Eq. (10), after extraction of the terminal optimum and full printed token. Unrounded 5% sensitivity changes no label on the 31 successful runs. Parse for Python means code extraction; a separate syntax check of all archived code passes but is not a semantic test. PaMOP's absolute-10^{-6} aggregate is withheld: exact equality of rounded display text is insufficient precision evidence. A conservative uncertainty of one last displayed unit yields two robust disagreements and nine unresolved comparisons among 11 available executed references; the two no-reference cases are separate. We do not report a selective 0/2 accuracy or infer a new exact-agreement fraction.

Table S12 Actual historical implementation and configuration, not original-author performance reproduction.

System	Model/implementation	Settings and actual solver
PaMOP	Project-maintained reconstruction; Azure GPT-5.4	$T = 0.2$; decomposition; at most 5 correction iterations; AMPL/Gurobi
Zero-shot	Azure GPT-5.4 fixed prompt	$T = 0$; one generation; 8192-token cap; gurobipy
OptMATH	Adapted released Qwen2.5-7B checkpoint	$T = 0.8$; sampling; seed 0; 8192-token cap; gurobipy
ORLM	Adapted released LLaMA-3-8B checkpoint	Greedy; seed 0; 8192-token cap; coptpy unavailable

Table S13 Noninterchangeable failure and assessment stages from archived records.

System	Stage/outcome	Count
PaMOP	Successful without / after correction	9 / 4
PaMOP	Final parse failure	5
PaMOP	Executed: robust discrepancy / precision unresolved / no reference	2 / 9 / 2
OptMATH	Unsuccessful process / objective disagreement among comparable executions	3 / 4
Zero-shot	Unsuccessful process / objective disagreement among comparable executions	2 / 4
ORLM	Execution environment unavailable	18

S4.2 Per-Case Outcomes and Precision Boundaries

P denotes final parse failure, E an unsuccessful process, N an unavailable execution environment, Y+ a classifiable objective match, Y– a classifiable discrepancy, Y? a successful run without a numerical reference, and U an unresolved absolute-precision comparison. U is neither a failed model nor a confirmed agreement. The historical five false feasibility flags on failed Python processes are excluded. Each source record and its hash is identified in the local versioned correction ledger.

For PaMOP case 254 the retained generated LP minimizes $100x + 72y$ subject to $60x + 40y \geq 10000$, $x \geq (x + y)/3$ and nonnegative variables. Since $100x + 72y \geq (5/3)(60x + 40y) \geq 50000/3$, with equality at $x = 500/3, y = 0$, the mathematical model has objective $50000/3$. This algebraic witness explains the display-rounding artifact but is not a recovered full-precision historical solver return or a complete source-semantic validation. The printed summary 16666.66667 is still insufficient at absolute 10^{-6}, so the case remains U in the machine-measurement ledger.

The archived comparison contains no evaluated common-18 generation for DeepOR or OR-R1. Their historical August 2026 availability context is retained, with no fabricated zero accuracy. Different prompts, adaptations, sampling and correction budgets,

Table S14 Common-18 corrected dispositions from archived records; no regenerated programs or solves.

Problem ID	PaMOP	OptMATH	ORLM	Zero-shot
14	Y–	Y+	N	Y+
23	U	Y+	N	Y+
34	U	Y+	N	Y+
59	U	Y–	N	Y–
69	U	Y–	N	Y–
72	U	Y+	N	Y+
84	P	Y+	N	Y+
88	U	Y+	N	Y+
96	Y–	Y–	N	Y–
117	P	Y–	N	Y+
190	P	Y+	N	Y+
202	U	Y+	N	Y+
208	U	Y+	N	Y+
219	Y?	E	N	E
232	P	E	N	Y–
237	Y?	E	N	E
254	U	Y+	N	Y+
262	P	Y+	N	Y+

reference predicates and execution availability prevent a provider/model ranking. No scalar-only readiness score is entered as executable-formulation performance.

S5 Closed Slot-Participation Audit

The G1–G3 audits were bounded source/record investigations and produced no experimental predictions. G1 found missing validated autonomous slot discovery; G2 distinguished slot participation from mention-candidate filtering. For G3, let $P_{\text{scalar}}(\hat{s})$ select only metadata entries whose declared shape is empty. It compares $K_G \cap P_{\text{scalar}}$ against P_{scalar} using saved TF–IDF IDs, without running a decoder.

Table S15 Static G3 inventory accounting on 331 saved TF–IDF records; these are set counts, not performance outcomes.

Measure	Count
Equal /different participation sets	301 /30
Additional schema-slot occurrences	134
Gold slots absent from predicted scalar metadata	172 /1847
Changed cases missing required gold keys in both conditions	29
Changed case without numeric scalar references	1
Cases where strict metadata-scalar A differs from original fallback A	7

The original implementation intersects K_G with all metadata keys, falling back to reference parameter keys; it does not first restrict shapes. Its saved slot counts sum to 1705, whereas strict metadata scalars give 1675, differing by 30 slots in seven cases. These baseline definitions cannot be interchanged. The changed cases add 134 occurrences: 132 keys are unannotated in the query reference, and two have symbolic string values. Their correctness is unknown, not automatically false. Whole-key success cannot distinguish the two conditions in 29 changed cases because both lack required keys; the remaining case cannot certify numeric scalar success. Equal-set cases supply no intervention under identical deterministic ordering. This is structural identifiability, not a measured zero effect or evidence of equivalence.

Only query 102 is both a changed case, a schema hit and a source of shared numeric targets (three). Its metadata/reference names differ and existing percentage conventions are inconsistent. Wrong-schema shared names do not certify shared semantics. Added slots could compete for mentions under no reuse, but that mechanical possibility does not supply an informative performance study here. All 49 prior primary factorial cases have equal sets. No 10–20-case preflight or new 331-case run was executed or recommended. The closed audit preserves per-case inventories, hashes and exact denominators; no algorithm or reference repair was applied.

S6 Relationship-Labeling Evidence and Recovery

S6.1 Provenance, Requests and Denominators

The packet contains 217 unique occurrence–declaration pairs in 16 source problems, all from NL4Opt train (713 source records), pinned to revision `49f1e0d66b7fdcd33305a7f281c2a7c13f5620ea`. The complete source-file SHA256 is

recorded with the source-to-result manifest. Selection is a high-priority convenience handoff with category coverage from an exposed 100-problem review packet, not a new probability sample or 217 independently sampled observations. Relationship IDs, problem IDs, source hashes and every selected object are retained. Reused mentions or declarations remain separate relationships. There are no missing or duplicate targets in the original 217-target frame for any model; historical replacements are additional attempts at a subset, not additional independent observations.

All retained study calls are dated October 27, 2026 (UTC). Original raw-response timestamps span 20:58:09–21:12:23; targeted-recovery responses span 21:45:10–21:52:12. Synthetic access probes and model-list lookups in preflight are not study judgments. Vertex uses the v1 `generateContent` route (project ID removed from release copies), io.net uses https://api.intelligence.io.solutions/api/v1/chat/completions, and Cohere uses https://api.cohere.com/v2/chat. Requested IDs and settings appear in main Section 6.3. Gemini and io.net responses echo their requested model strings; Cohere does not supply a separately authenticated version in the retained response. No serving weight checksum is available. Unrecorded top-p, inference seeds and other defaults are unknown.

Table S16 Retained HTTP attempts, distinct from model-item judgments. A response with HTTP 200 may still contain invalid judgment objects. The initial Cohere recovery transport was rejected before the amended native-schema transport.

Service	Original attempts	Original HTTP 200	Recovery attempts	Recovery HTTP 200
Gemini	33	33	0	0
io.net/DeepSeek	33	33	3	3
Cohere	33	32	58	27

The remaining original Cohere attempt returns HTTP 422. Recovery retains 27 Cohere HTTP 400 responses from the initial native-schema format, four HTTP 422 responses, and 27 HTTP 200 responses after amendment. These 58 attempts must not be described as 27 total requests. The completed replacement batches target four io.net items and 88 Cohere items. Neither HTTP retries nor transport amendments constitute new target observations. The amended schema removes unsupported native length constraints and supplies explicit enum types; local validation is unchanged. This is a historical protocol deviation, fully retained, not an M5 rerun.

The original response contract requires exactly one object per requested ID and ten named fields: item ID, decision, explanation, evidence, variable IDs, constraint IDs, missing relations, missing implicit objective terms, shared-mention ambiguity and uncertainty. Evidence contains literal contiguous source quotes and supporting/contradicting/context tags. A unique complete envelope may be extracted from prose, an alias may be mapped only via the source's explicit alias dictionary, and structured omission-note objects may be serialized with an exact JSON round trip.

No decision or quotation is rewritten. These deterministic rules repair 59 originally invalid objects; 500 originally valid objects remain unchanged. The final set adds 72 valid historical replacements and retains 20 invalid ones. Structurally invalid labels are not counted as a fifth nominal category or scored as semantic errors.

Table S17 Valid-label distributions after Recovery v2.1. C = CORRECT, I = INCORRECT, A = AMBIGUOUS, CD = CANNOT_DETERMINE; these are model assertions, not human reference labels. Invalid judgments are excluded from these label totals.

Service	C	I	A	CD	Valid total
Gemini	134	82	0	1	217
io.net/DeepSeek	130	84	2	1	217
Cohere	177	20	0	0	197

For $N = 197$ complete items and $n = 3$ raters, let n_{ic} count judgments in nominal category c. We recompute

$$\bar{P} = \frac{1}{N}\sum_i \frac{\sum_c n_{ic}(n_{ic}-1)}{n(n-1)}, \quad P_e = \sum_c \left(\frac{\sum_i n_{ic}}{Nn}\right)^2, \quad \kappa = \frac{\bar{P}-P_e}{1-P_e}. \tag{S2}$$

Using all four categories gives $\bar{P} = 0.7394247039$, $P_e = 0.5858578050$ and $\kappa = 0.3708071786$. Complete-case selection is conditional on output validity and may be informative. Offline-only repair yields 128 complete items and $\kappa = 0.4293652076$; the original 89 complete items give 0.5016425265, unchanged when those exact 89 items are assessed in the final records. No interval or inferential model treats the 217 relationships as independent. The residual 20 items are all final Cohere nonliteral-evidence failures, individually listed in the machine-readable `llm217_missing20.csv`. The complete-case partition $119 + 77 + 1 = 197$, together with these 20, exhausts the frame of 217.

S6.2 Identified Protocol and Disagreement Witnesses

These examples are traceable observations, not human reference adjudications. Item `00330c75d90c5e1b24b115da`, problem −790646361, originally uses Cohere variable ID `spoon` where the supplied variable is `spoons`. The explicit source alias permits deterministic repair without changing the decision. Item `32f85abf70d7e3065d1ef01f`, problem −2101760318, retains a Cohere quote "You must consume at least 50 units of sugar" that is not a literal substring of that problem's source. It remains invalid; the numerical decision is not adjudicated by this string check.

For item `a8ccbba13aa191a89f74a58b`, problem −790646361, the three valid labels are CORRECT/INCORRECT/CORRECT (Gemini/io.net/Cohere). The explanations conflict over whether the targeted repeated 1.5 occurrence is a steel or

rubber quantity. Item `62b5c72b04145f4b57de8ef9`, problem 1470713712, has the same label pattern, with competing explanations about the 40-pound resource bound and the supplied coefficients. These are examples of equal-value/resource and declaration-scope disagreement, not evidence that one service is generally right. Item `9daac1d92d950356baeeb7ed`, problem −1202959445, has three CORRECT labels but no lexical occurrence endpoint. The protocol classifies it UNRESOLVED, explaining why raw label unanimity (120) exceeds determinate unanimity (119).

Table S18 Per-problem relationship counts, complete three-model cases and final dispositions. These 16 clusters, rather than 217 independent problems, define the source grouping. U denotes determinate unanimity, D disagreement, R unresolved and I invalid/incomplete.

Problem ID	Items	Complete	U	D	R	I
-1045601007	13	12	10	2	0	1
-1119351614	13	13	10	3	0	0
-1202959445	6	6	5	0	1	0
-1734964325	1	1	0	1	0	0
-1761770632	14	12	7	5	0	2
-2101760318	16	15	8	7	0	1
-2134084888	29	23	9	14	0	6
-214665832	21	21	7	14	0	0
-337748765	8	8	8	0	0	0
-655808311	10	8	6	2	0	2
-790646361	21	17	9	8	0	4
1405331961	10	10	8	2	0	0
1470713712	10	9	1	8	0	1
1482174936	14	12	12	0	0	2
603259629	15	15	11	4	0	0
781521487	16	15	8	7	0	1
Total	217	197	119	77	1	20

The local release candidate's 651-row derived ledger identifies each model/item, original/offline/final validity, label, origin, final failure and cluster. Frozen raw responses, prompts, request configurations and original/recovery hashes support independent verification. Public copies redact account/project and upload-location metadata while preserving scientific content and recording every transformed hash. No human-verification ledger contains accepted labels. The audit supplies no semantic precision, recall, model superiority, autonomous-system validation or approved training data.

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